

Problem

Oscilloscope measurements indicate that the peak voltage across a load and the peak current through it are, respectively, $210\angle 60^\circ \text{ V}$ and $8\angle 25^\circ \text{ A}$. Determine:

(a) the real power

(b) the apparent power

(c) the reactive power

(d) the power factor

Step-by-step solution

Step 1 of 4

(a)

$$\bar{S} = \frac{1}{2} V I^*$$

$$\bar{S} = \frac{1}{2} (210 \angle 60^\circ) (8 \angle -25^\circ)$$

$$\bar{S} = 840 \angle 35^\circ$$

$$P = S \cos \theta$$

$$P = 840 \cos 35^\circ$$

$$\boxed{P = 688.1 \text{ W}}$$

Step 2 of 4

(b)

$$\boxed{S = 840 \text{ VA}}$$

Step 3 of 4

(c)

$$Q = S \sin \theta$$

$$Q = 840 \sin 35^\circ$$

$$\boxed{Q = 481.8 \text{ VAR}}$$

Step 4 of 4

(d)

$$\text{P f} = \frac{P}{S}$$

$$\text{P f} = \cos 35^\circ$$

$$\boxed{\text{P f} = 0.8191} \quad [\text{Lagging}]$$